Face Detection

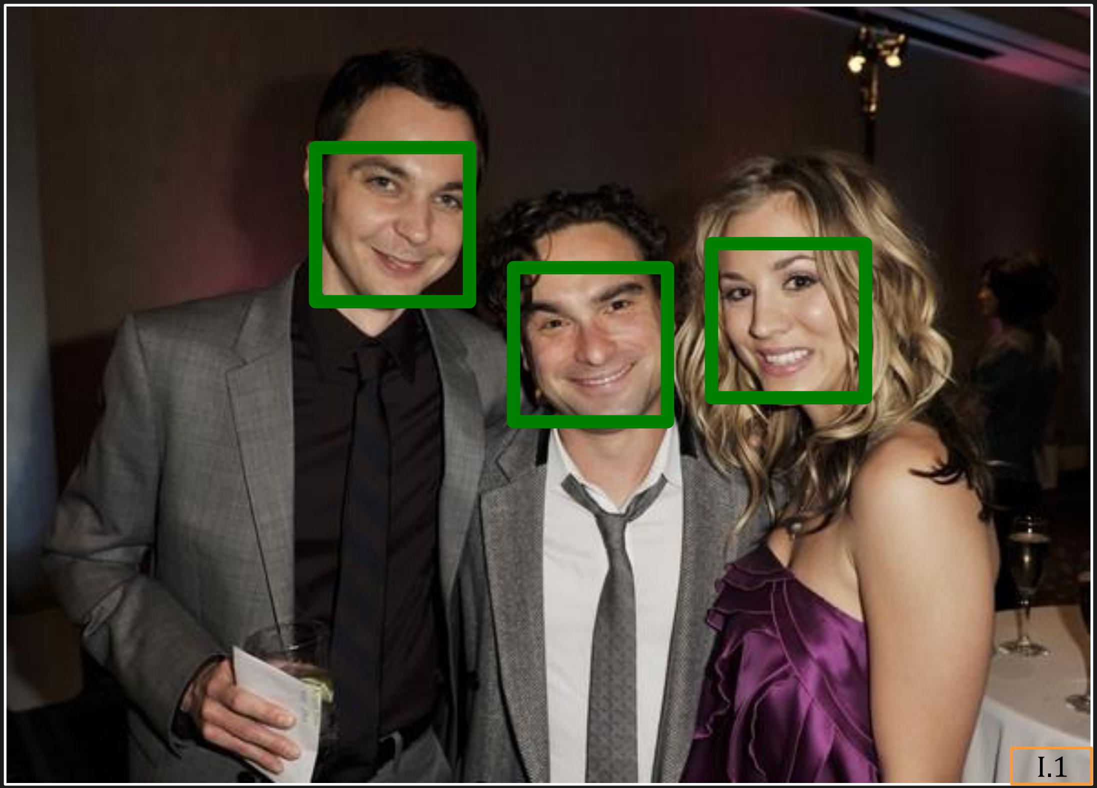
Computer Vision: CS 566

Computer Science

University of Wisconsin-Madison

What is Face Detection?

Locate human faces in images



Face Detection

Locate human faces in images.

Topics:

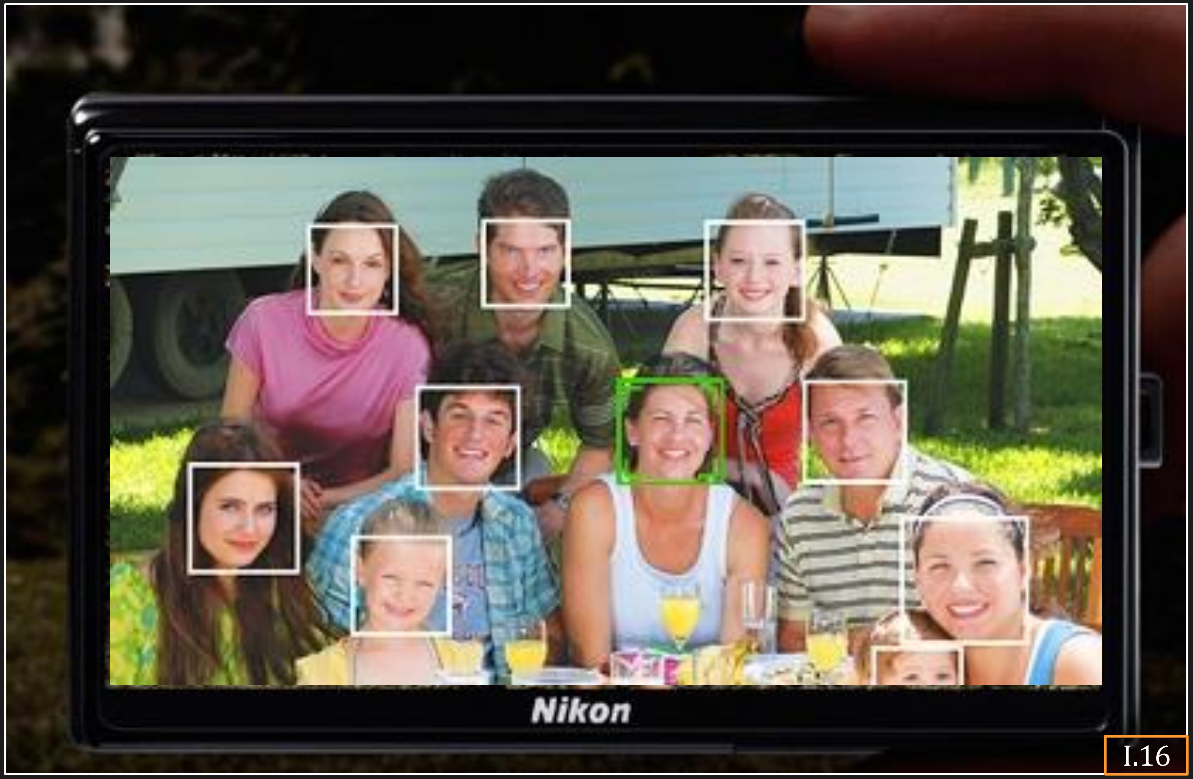
(1) Features for Face Detection

(2) Integral Images

(3) Nearest Neighbor Classifier

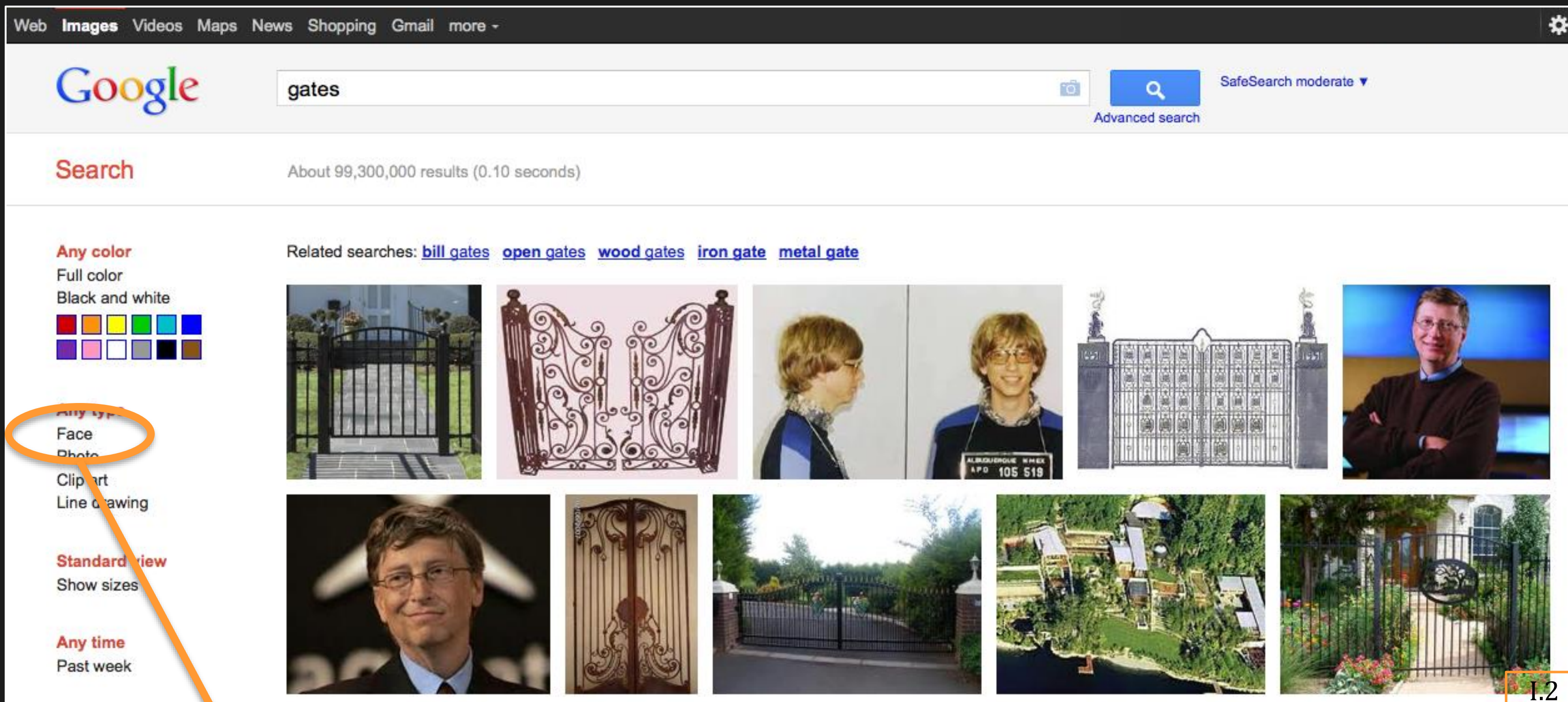
(4) Support Vector Machine

Where is Face Detection Used?



Cameras (for Autofocus)

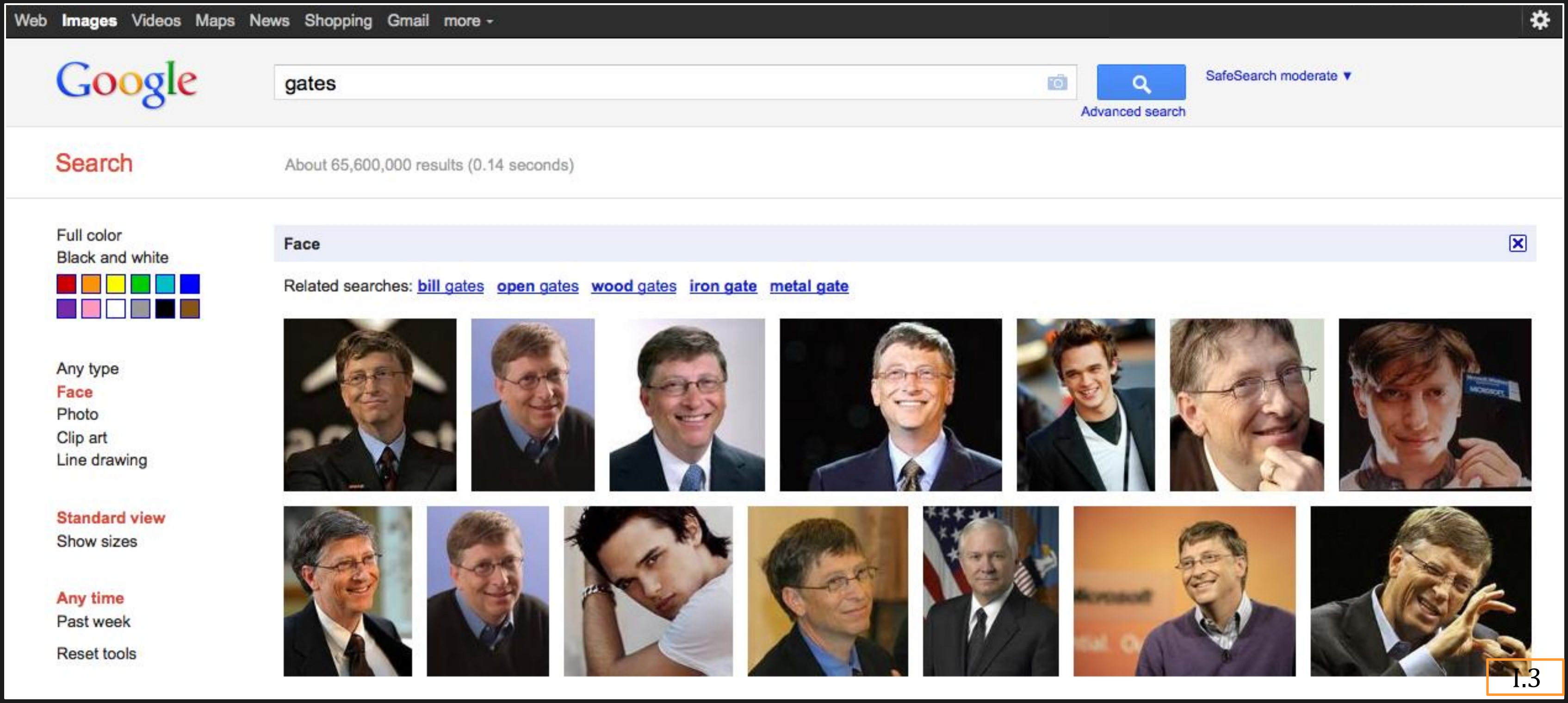
Where is Face Detection Used?



Face Detection

Search Engines

Where is Face Detection Used?



Only Faces of Gates now

Search Engines

Where is Face Detection Used?

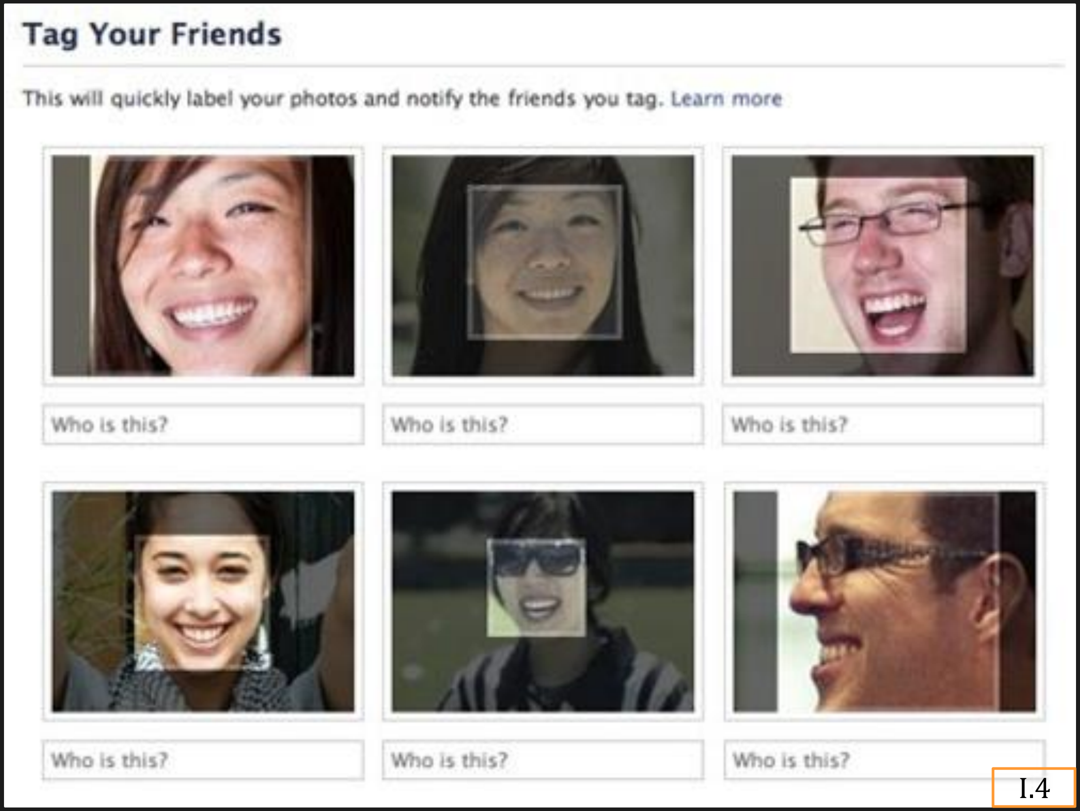
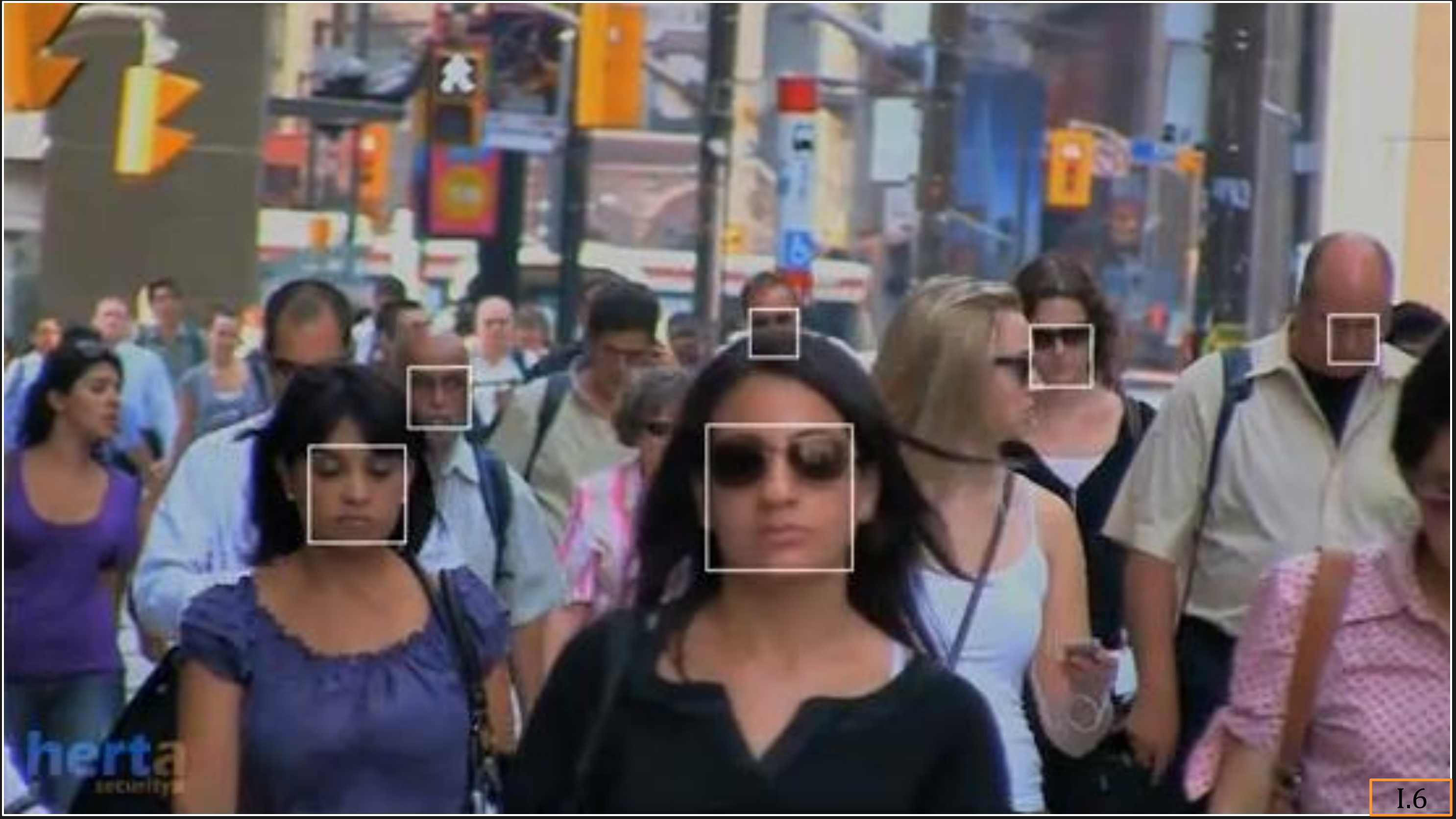


Photo Tagging in Social Networks

Where is Face Detection Used?



Surveillance and Monitoring

How Humans Detect Faces?

We do not know yet!

Some Conjectures:

- Memory prediction model

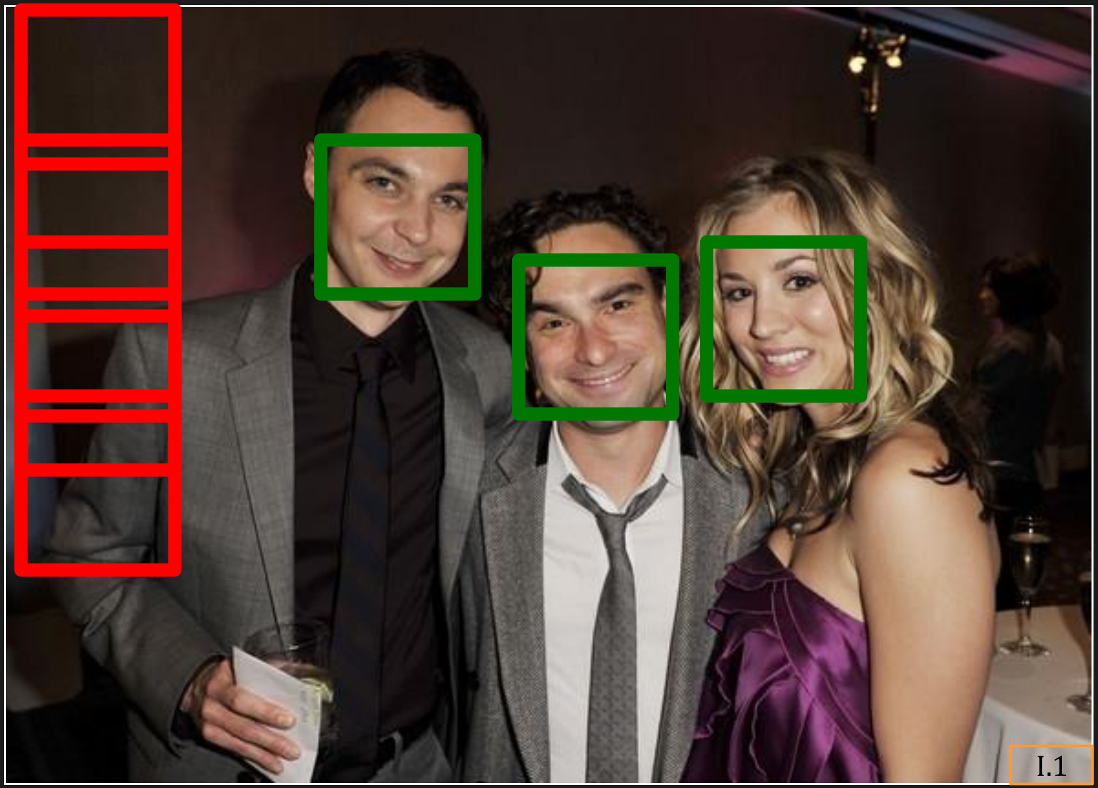
  Match image with face description (model) in memory.

- Parallel computing

  Detect faces at multiple location/scale combination.

Face Detection in Computers

Slide windows of different sizes across image.
At each location match window to face model.



Face Detection Framework

For each window

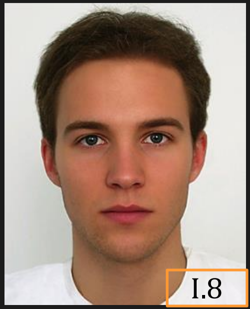
Extract Features → $\begin{bmatrix} \mathbf{f} \end{bmatrix}$ → Match Face Model → Yes / No

Features: Which features represent faces well?

Classifier: How to construct a face model and efficiently classify features as face or not?

Characteristics of Good Features

Discriminate Face/Non-Face



$\neq$

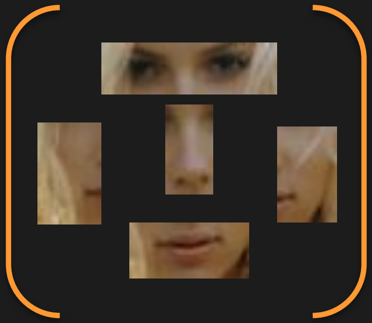
Extremely Fast to Compute

Need to evaluate tens of thousands windows in an image.

What are Good Features?

Interest Points (SIFT, Corners, Edges)?

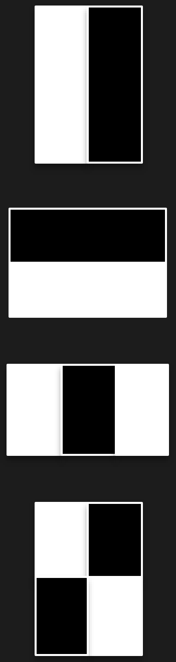
Facial Components?



Haar Features

Set of Correlation Responses to Haar Filters

Input Image

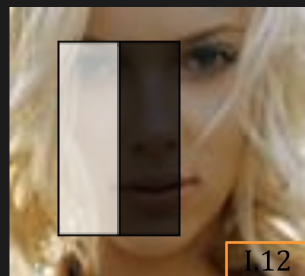


$$\otimes \begin{bmatrix} H_A \\ H_B \\ H_C \\ H_D \\ \vdots \end{bmatrix} = \begin{bmatrix} V_A[i,j] \\ V_B[i,j] \\ V_C[i,j] \\ V_D[i,j] \\ \vdots \end{bmatrix}$$

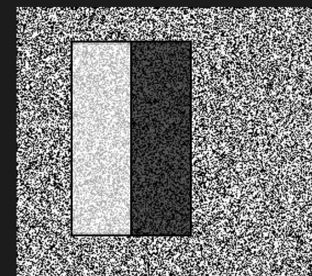
Haar Filters

Haar Features
$\mathbf{f}[i,j]$

Discriminative Ability of Haar Feature

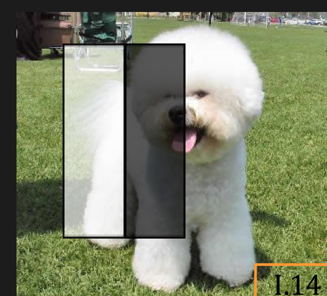


$V_A = 64$



$V_A \approx 0$

$V_A = 16$

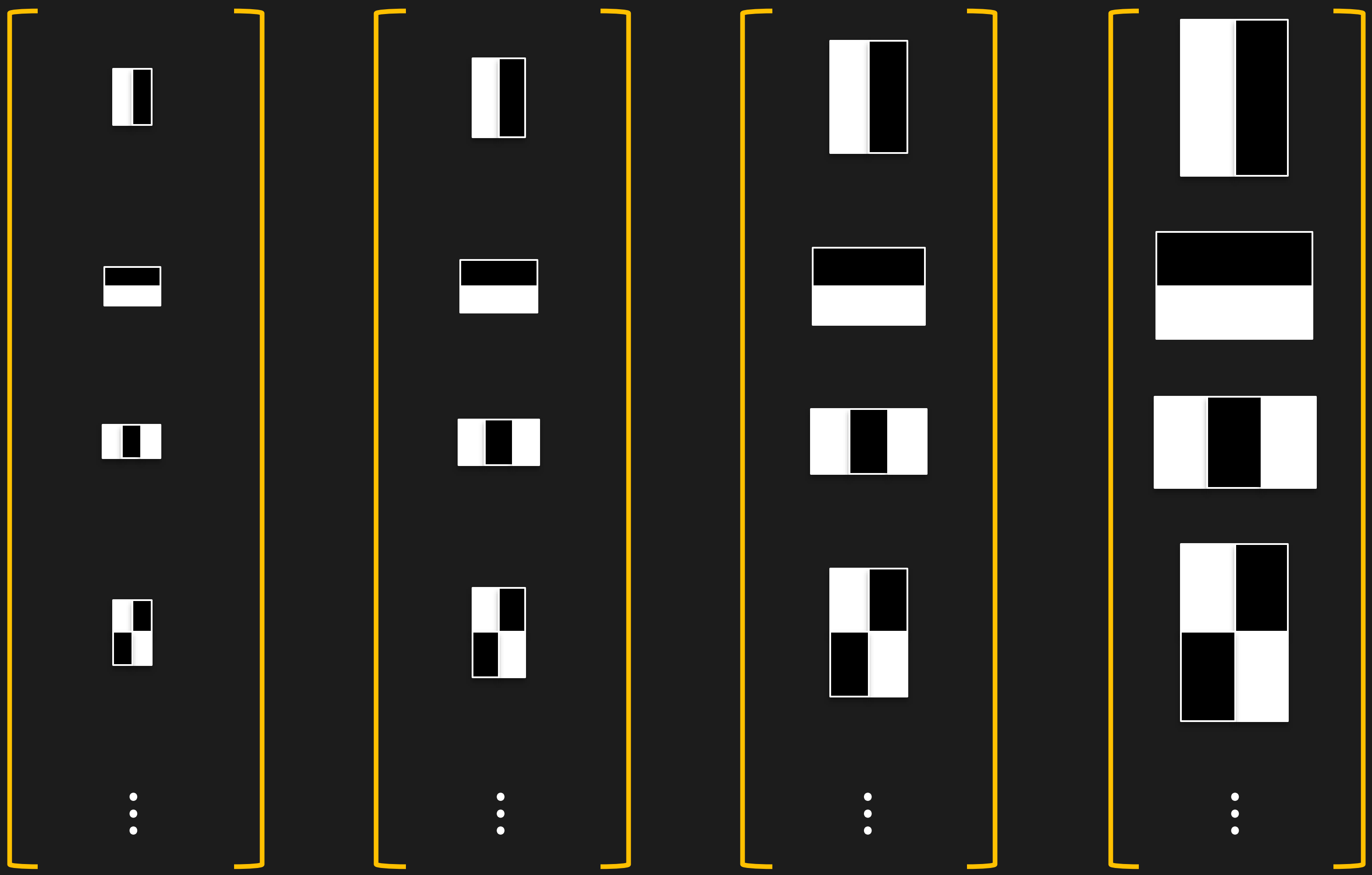


$V_A = -127$

Haar Features are Sensitive to Directionality of Patterns.

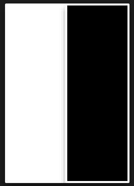
Detecting Faces of Different Size

Compute Haar Features at Different Scales to
Detect Faces of Corresponding Sizes.



Computing A Haar Feature

$\otimes$



H_A

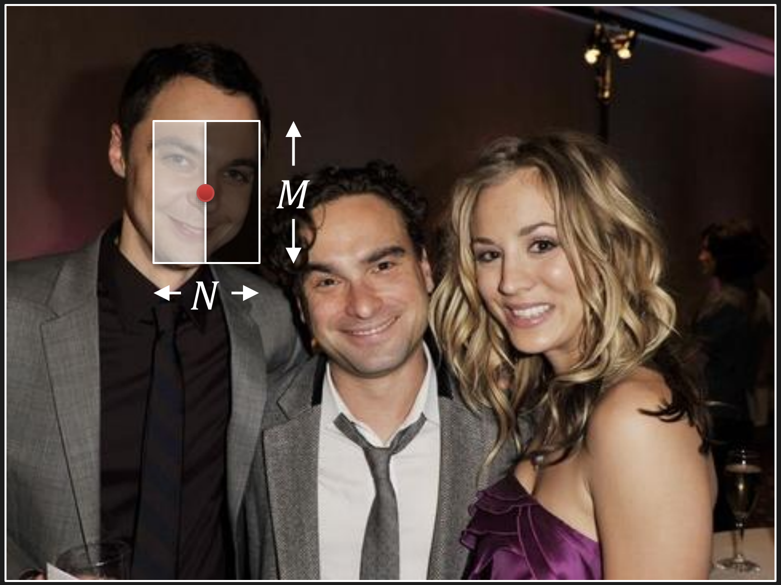
White = 1, Black = -1

Response to Filter H_A at location (i, j):

$$V_A[i,j] = \sum_m \sum_n I[i-m, j-n] H_A[m,n]$$

$$V_A[i,j] = \sum (\text{pixel intensities in white area}) - \sum (\text{pixels intensities in black area})$$

Haar Feature: Computation Cost



$$Value = \sum(pixel\ intensities\ in\ white) - \sum(pixel\ intensities\ in\ black)$$

Computation cost = $(N \times M - 1)$ additions per pixel per filter per scale

Can We Do Better?

Integral Image

A table that holds the sum of all pixel values to the left and top of a given pixel, inclusive.

98	110	121	125	122	129
99	110	120	116	116	129
97	109	124	111	123	134
98	112	132	108	123	133
97	113	147	108	125	142
95	111	168	122	130	137
96	104	172	130	126	130

Image I

98	208	329	454	576	705
197	417	658	899	1137	1395
294	623	988	1340	1701	2093
392	833	1330	1790	2274	2799
489	1043	1687	2255	2864	3531
584	1249	2061	2751	3490	4294
680	1449	2433	3253	4118	5052

Integral Image II

[Crow 1985, Viola 2002]

Integral Image

A table that holds the sum of all pixel values to the left and top of a given pixel, inclusive.

98	110	121	125	122	129
99	110	120	116	116	129
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Integral Image II

Summation Within a Rectangle

Fast summations of arbitrary rectangles using integral images

98	110	121	125	122	129
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584	1249	2061	2751	3490	4294
680	1449	2433	3253	4118	5052

P

Integral Image II

$$Sum = II_P + \cdots$$
$$= \ 3490 + \cdots$$

Summation Within a Rectangle

Fast summations of arbitrary rectangles using integral images

98	110	121	125	122	129
99	110	120	116	116	129
97	109	124	111	123	134
98	112	132	108	123	133
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Image I

98	208	329	454	576	705	
197	417	658	899	1137	1395	Q
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489	1043	1687	2255	2864	3531	
584	1249	2061	2751	3490	4294	P
680	1449	2433	3253	4118	5052	

Integral Image II

$$Sum = II_P - II_Q + \cdots$$
$$= 3490 - 1137 + \cdots$$

Summation Within a Rectangle

Fast summations of arbitrary rectangles using integral images

98	110	121	125	122	129
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97	109	124	111	123	134
98	112	132	108	123	133
97	113	147	108	125	142
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489	1043	1687	2255	2864	3531
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680	1449	2433	3253	4118	5052

Integral Image II

$$Sum = II_P - II_Q - II_S + \cdots$$

$$= 3490 - 1137 - 1249 + \cdots$$

Summation Within a Rectangle

Fast summations of arbitrary rectangles using integral images

98	110	121	125	122	129
99	110	120	116	116	129
97	109	124	111	123	134
98	112	132	108	123	133
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R Q S P

Integral Image II

$$Sum = II_P - II_Q - II_S + II_R$$

$$= 3490 - 1137 - 1249 + 417 = 1521$$

Computational Cost: Only 3 additions

Haar Response using Integral Image

98	110	121	125	122	129
99	110	120	116	116	129
97	109	124	111	123	134
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Integral Image II

$$V_A = \sum(pixels\ in\ white) - \sum(pixels\ in\ black)$$

Haar Response using Integral Image

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489	1043	1687	2255	2864	3531
584	1249	2061	2751	3490	4294
680	1449	2433	3253	4118	5052

Integral Image II

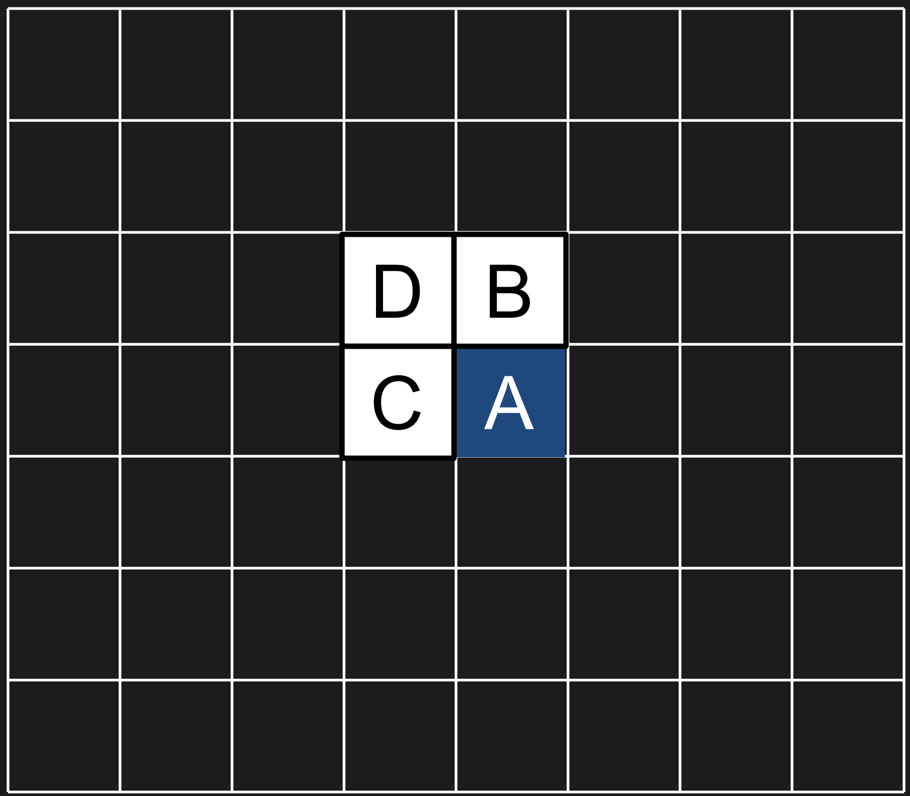
$$V_A = \sum(pixel\ intensities\ in\ white) - \sum(pixel\ intensities\ in\ black)$$

$$= (II_O - II_T + II_R - II_S) - \left(II_P - II_Q + II_T - II_O\right)$$

$$= (2061 - 329 + 98 - 584) - (3490 - 576 + 329 - 2061) = 64$$

Computational Cost: Only 7 additions

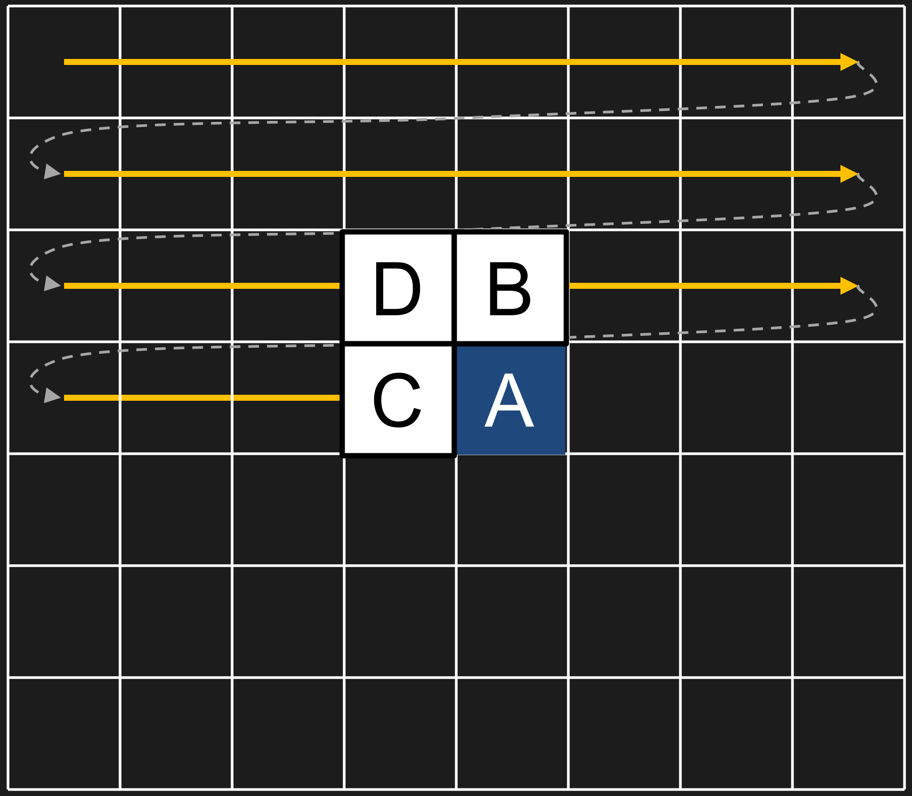
Computing Integral Image



Let I_A and II_A be the values of Image and Integral Image, respectively, at pixel A.

$$\boxed{II_A = II_B + II_C - II_D + I_A}$$

Computing Integral Image



Raster Scanning

Let I_A and II_A be the values of Image and Integral Image, respectively, at pixel A.

$$\boxed{II_A = II_B + II_C - II_D + I_A}$$

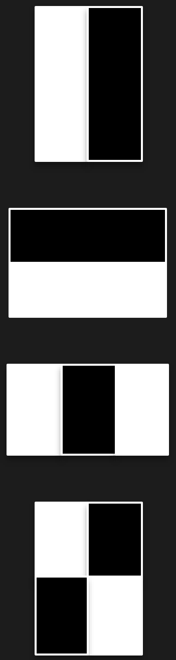
Haar Features Using Integral Images

Integral image needs to be computed once per test image.

Allows very fast computations of Haar features.

Input Image

$\otimes$



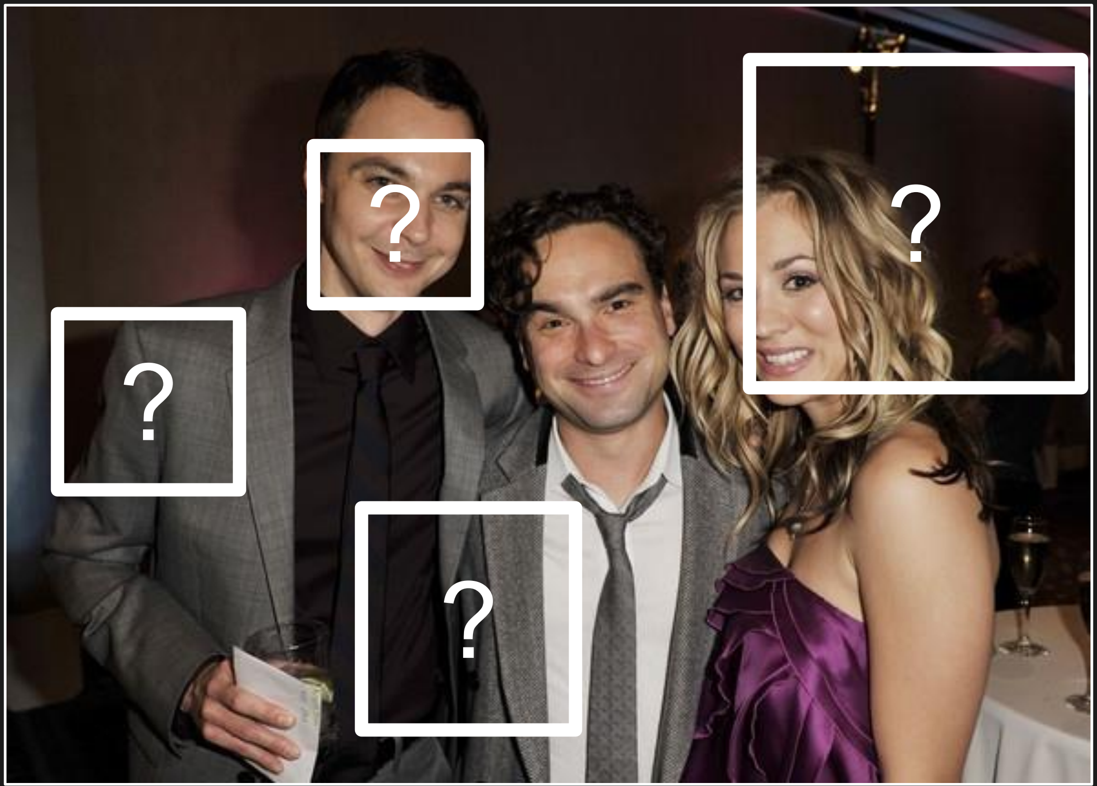
$$\begin{bmatrix} H_A \\ H_B \\ H_C \\ H_D \\ \vdots \end{bmatrix} = \begin{bmatrix} V_A[i,j] \\ V_B[i,j] \\ V_C[i,j] \\ V_D[i,j] \\ \vdots \end{bmatrix}$$

Haar Filters

Haar Features

Classifier for Face Detection

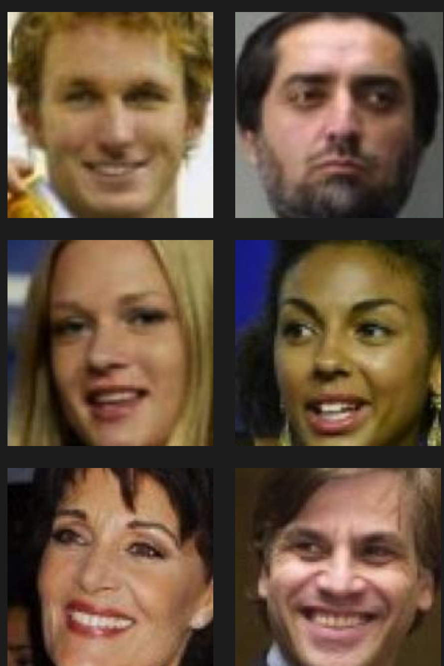
Given the features for a window, how to decide whether it contains a face or not?



Feature Space

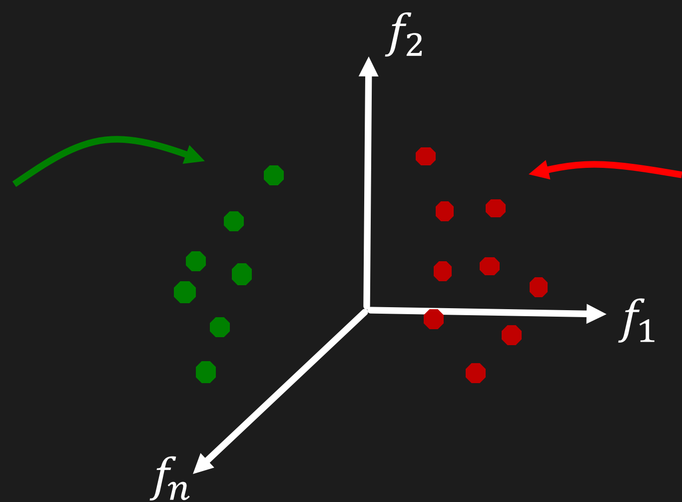
Haar features $\mathbf{f}$ (a vector) at a pixel

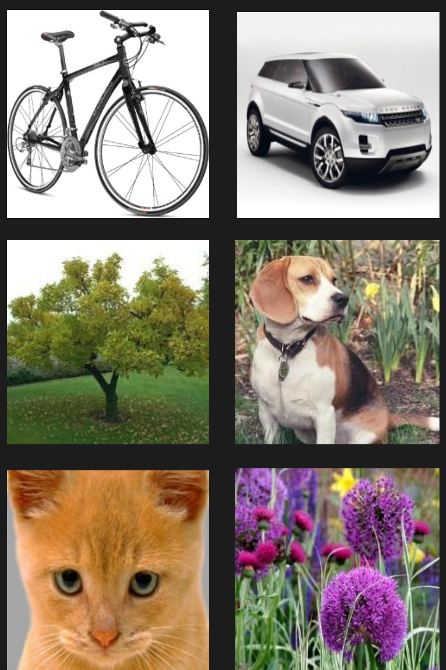
is a point in an n-D space, $\mathbf{f} \in \mathbb{R}^n$



Training Data

of Face



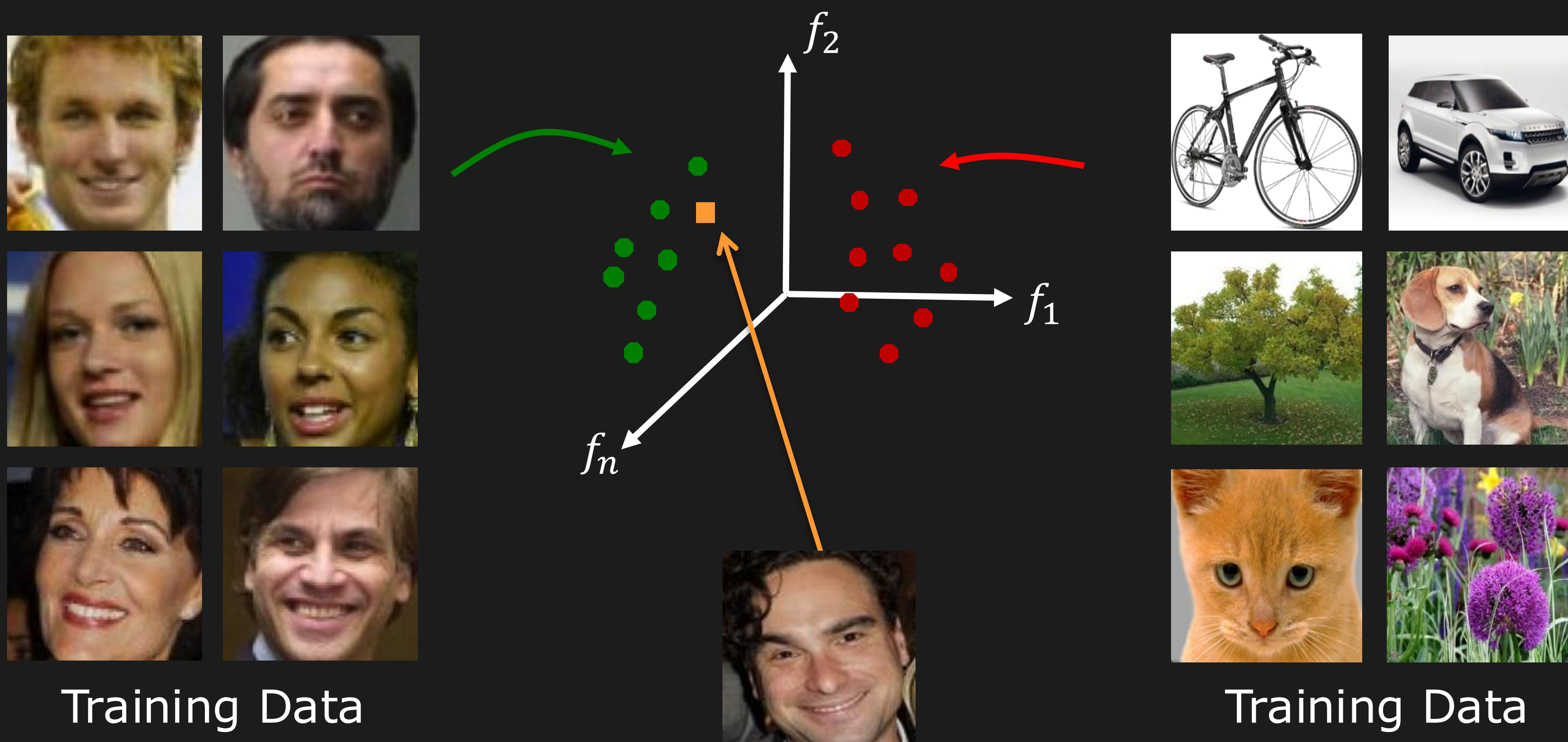


Training Data

of Non-Face

Classifier

Decision engine that classifies a test image as face or not



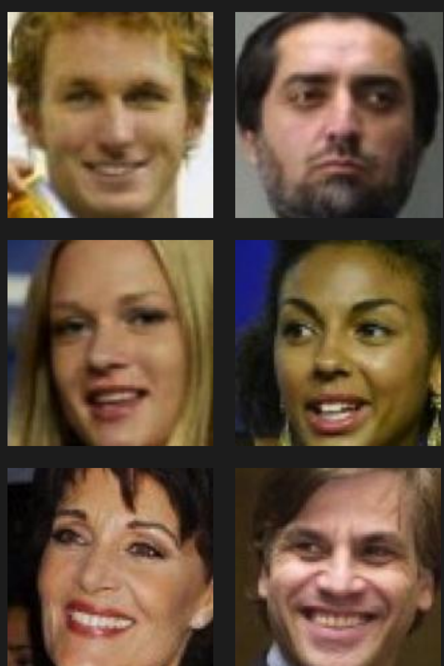
Training Data of Face

Test Image

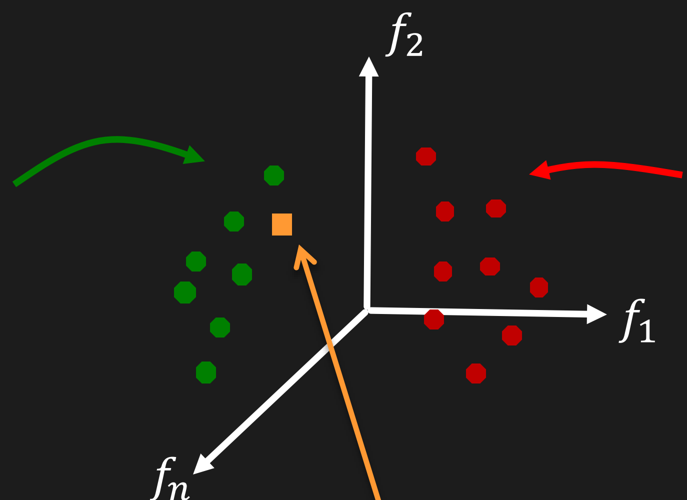
Training Data of Non-Face

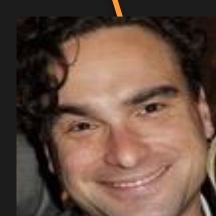
Nearest Neighbor Classifier

Find the nearest training sample using $\mathcal{L}^2$ distance and assign its label.

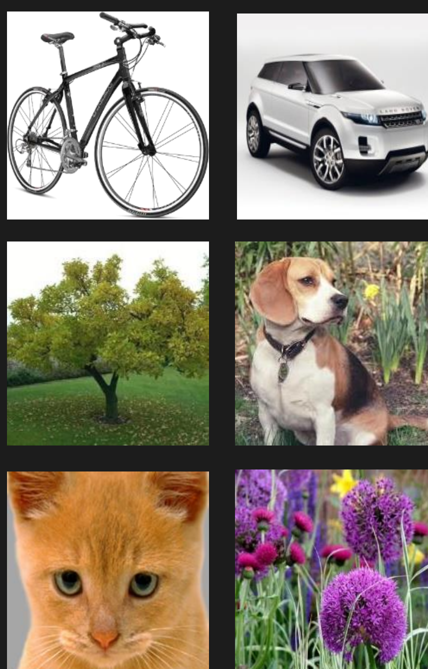


Training Data of Face





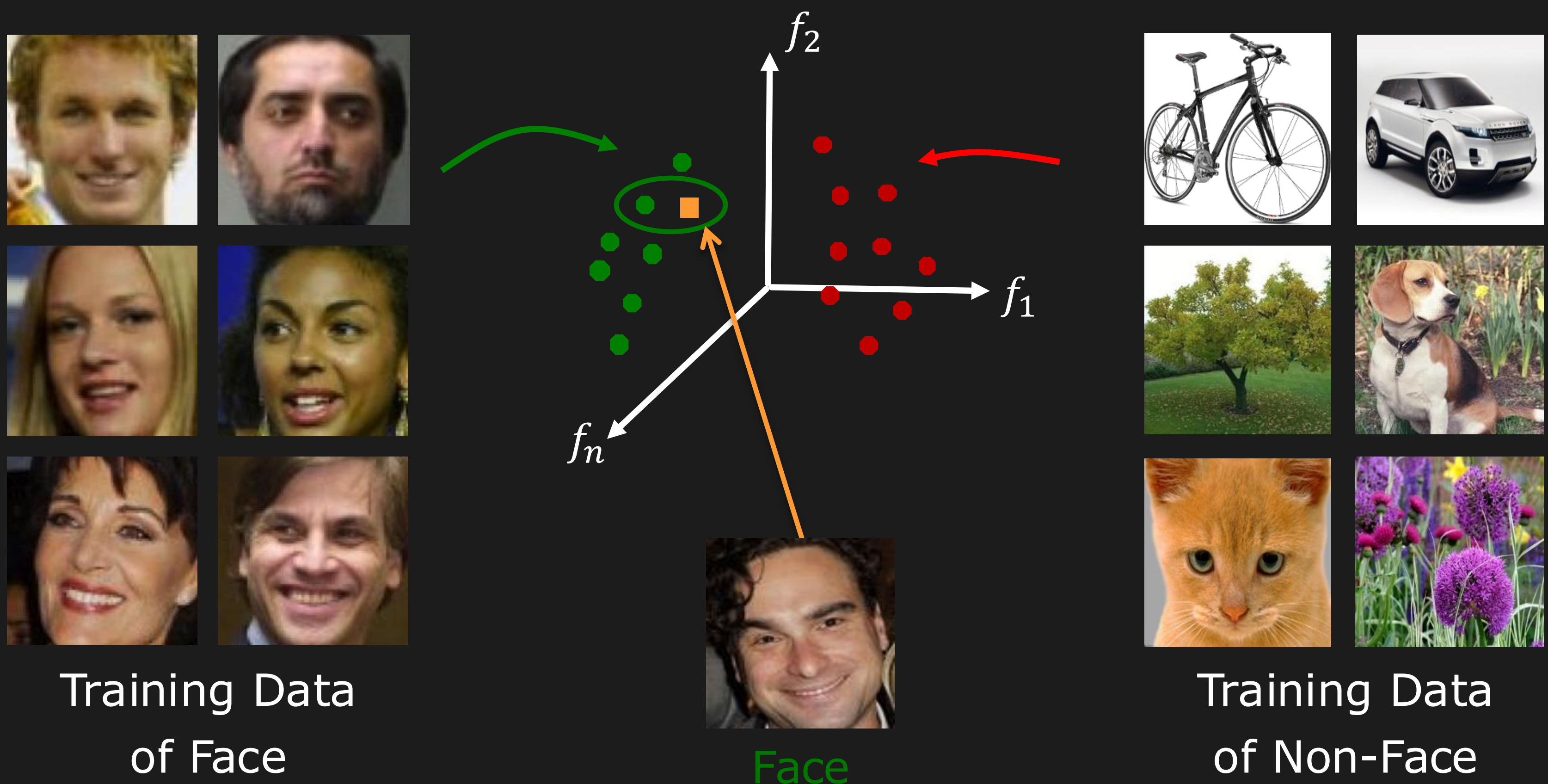
Test Image



Training Data of Non-Face

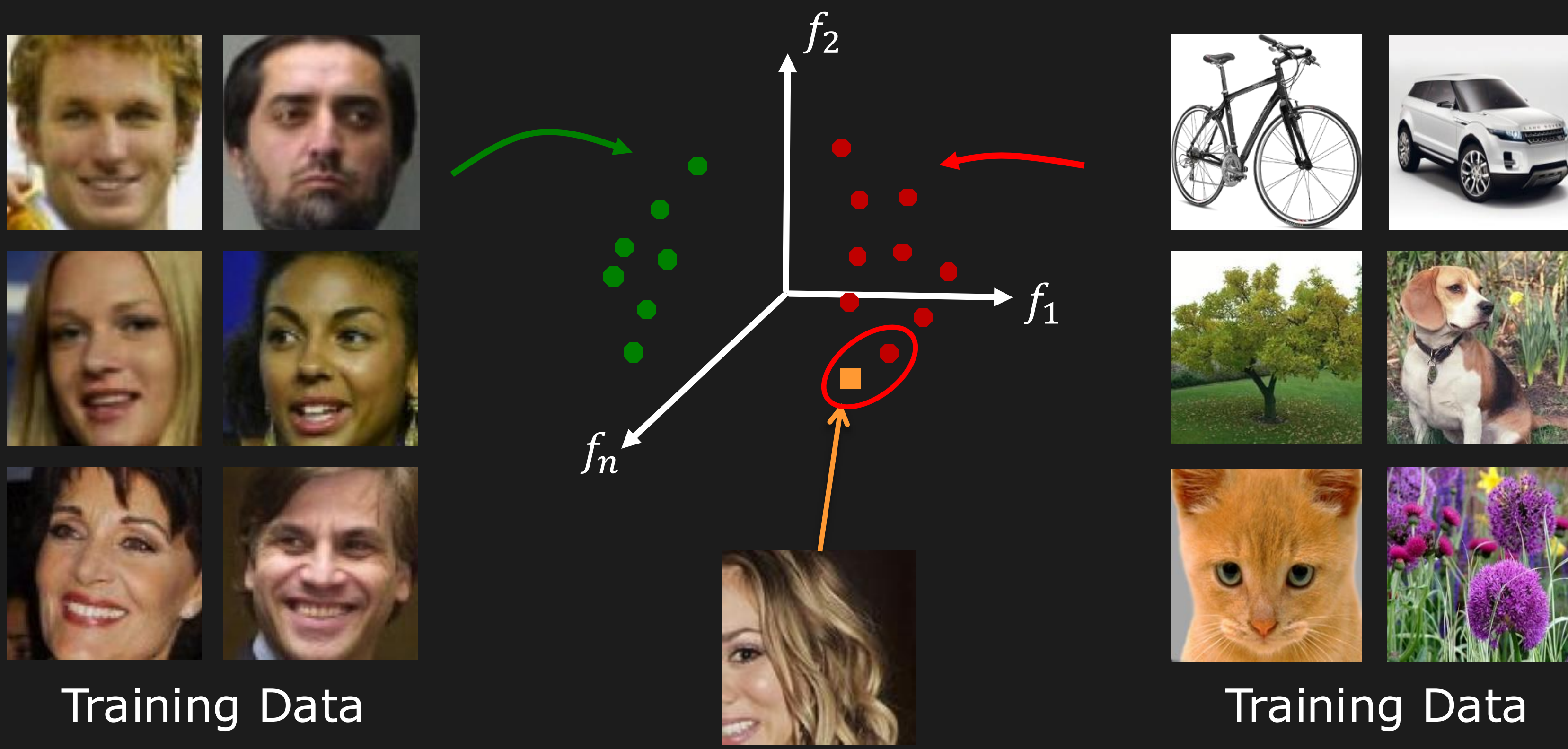
Nearest Neighbor Classifier

Find the nearest training sample using $\mathcal{L}^2$ distance and assign its label.



Nearest Neighbor Classifier

Find the nearest training sample using $\mathcal{L}^2$ distance and assign its label.



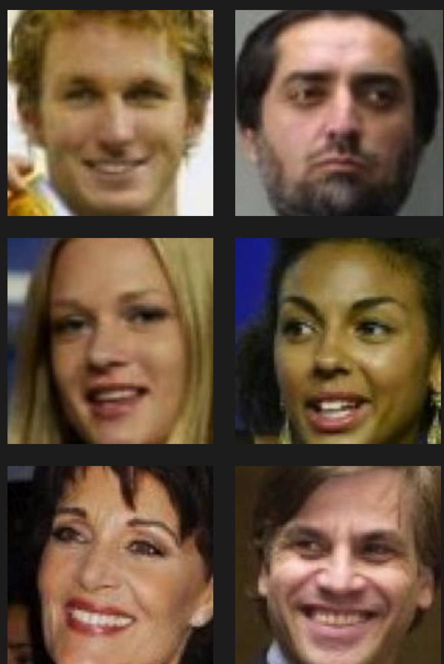
Training Data of Face

Not Face

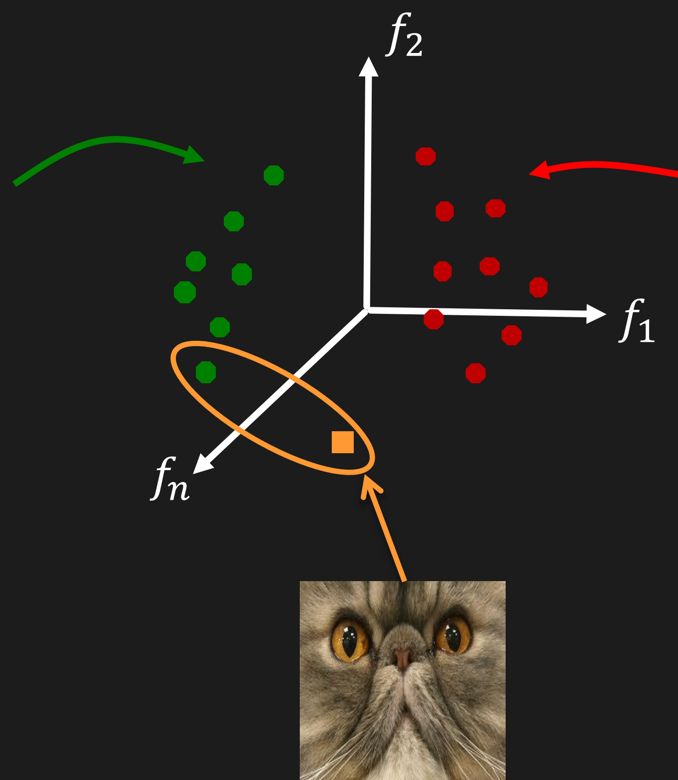
Training Data of Non-Face

Nearest Neighbor Classifier

Find the nearest training sample using $\mathcal{L}^2$ distance and assign its label.

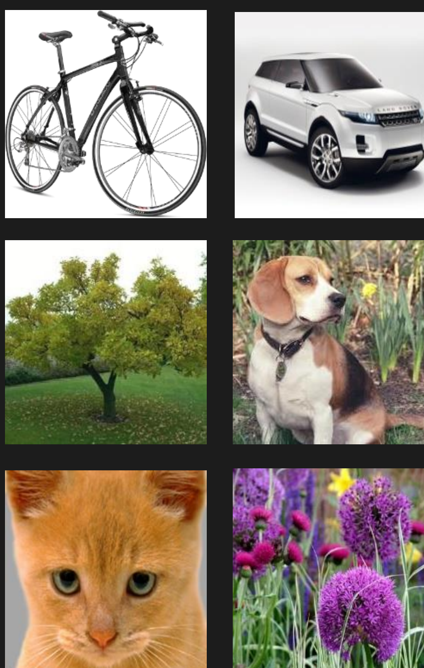


Training Data of Face



Face

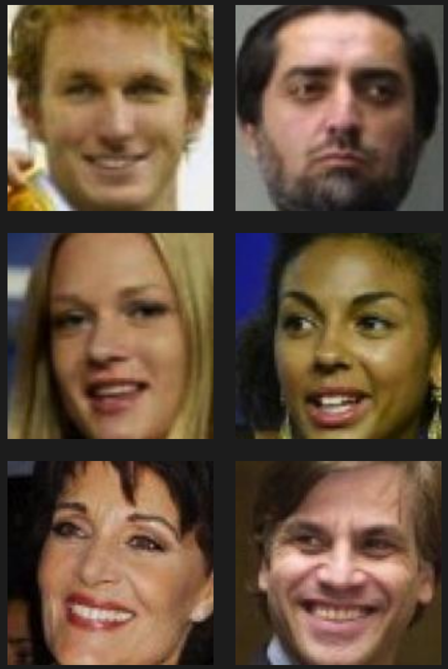
False Positive



Training Data of Non-Face

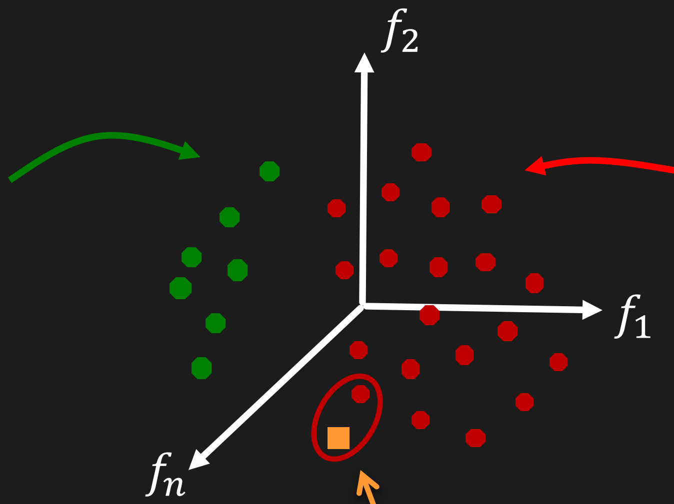
Nearest Neighbor Classifier

Larger the training set, more robust the NN classifier

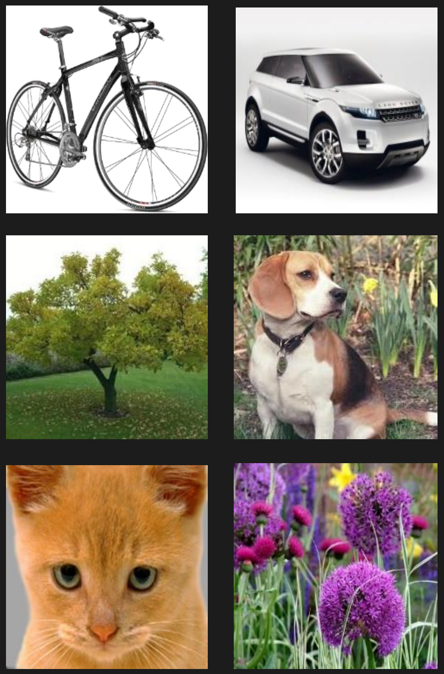


Training Data

of Face



Non-Face

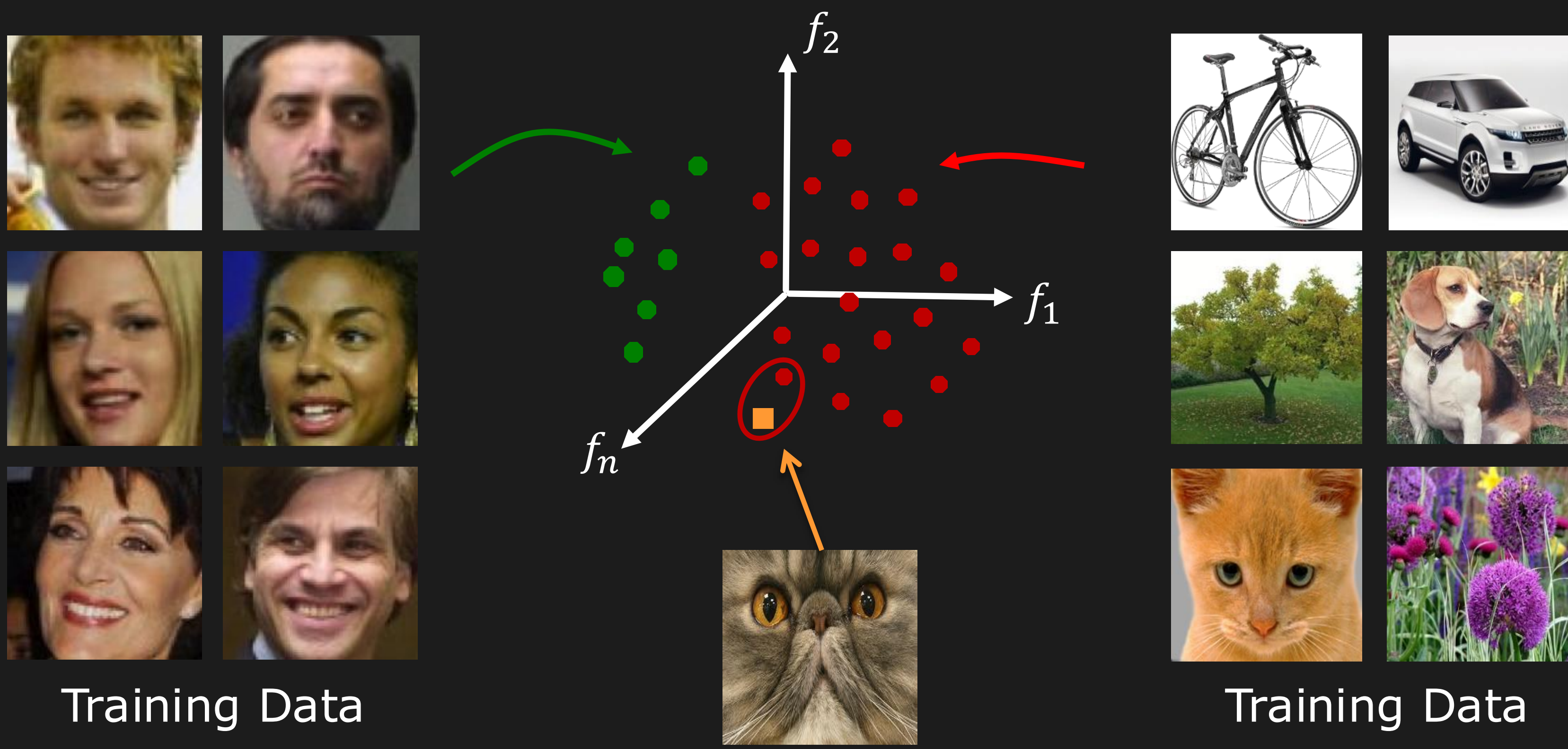


Training Data

of Non-Face

Nearest Neighbor Classifier

Larger the training set, slower the NN classifier



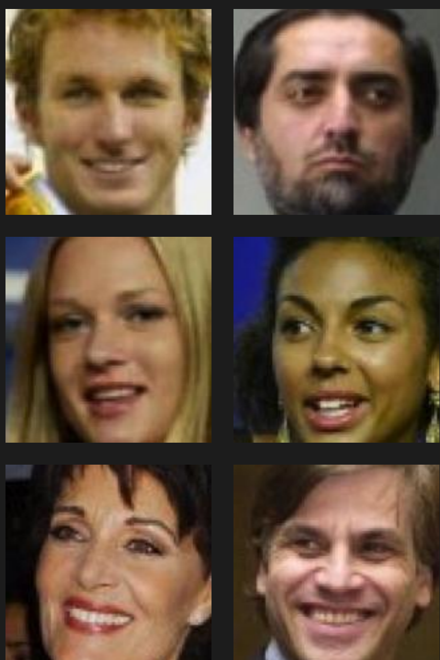
Training Data of Face

Non-Face

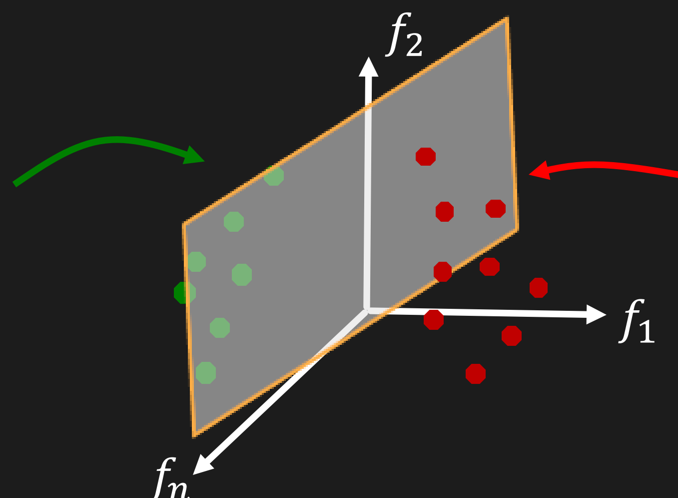
Training Data of Non-Face

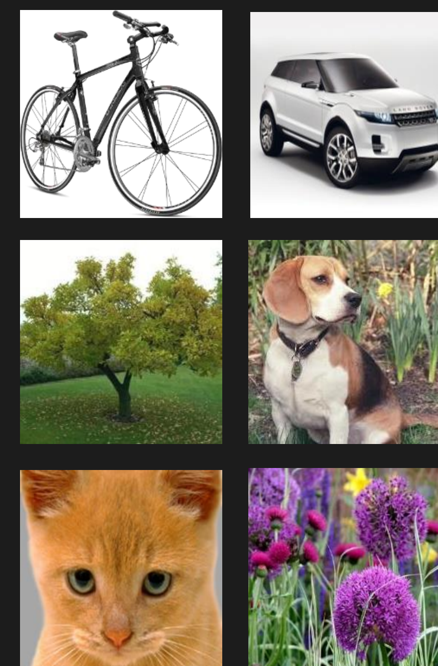
Decision Boundary

A simple decision boundary separating face and non-face classes will suffice.



Training Data of Face

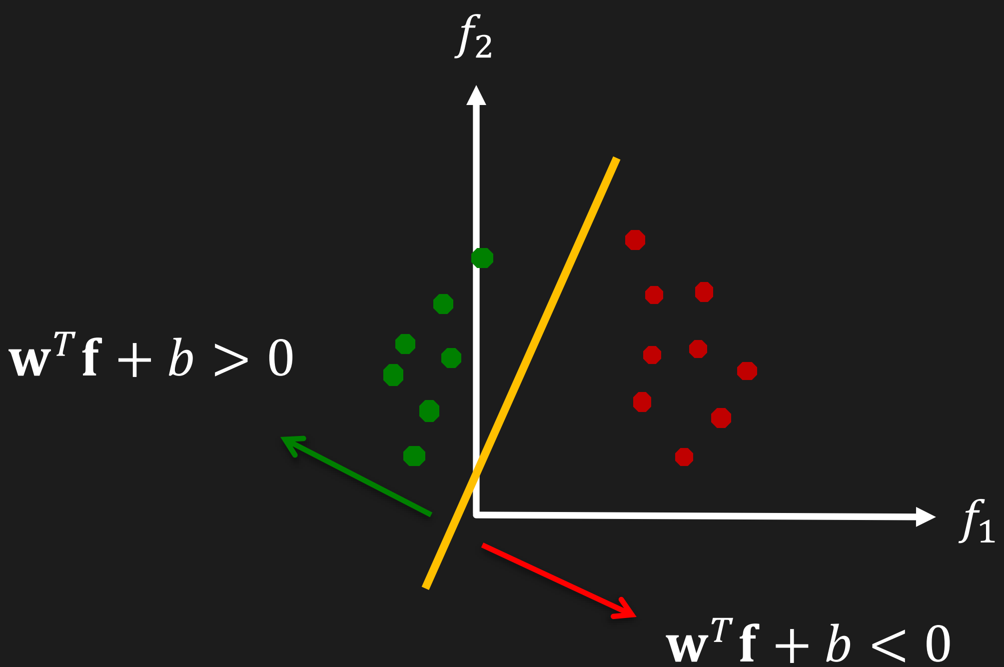




Training Data of Non-Face

Linear Decision Boundaries

A Linear Decision Boundary in 2-D space is a 1-D Line



Equation of Line:

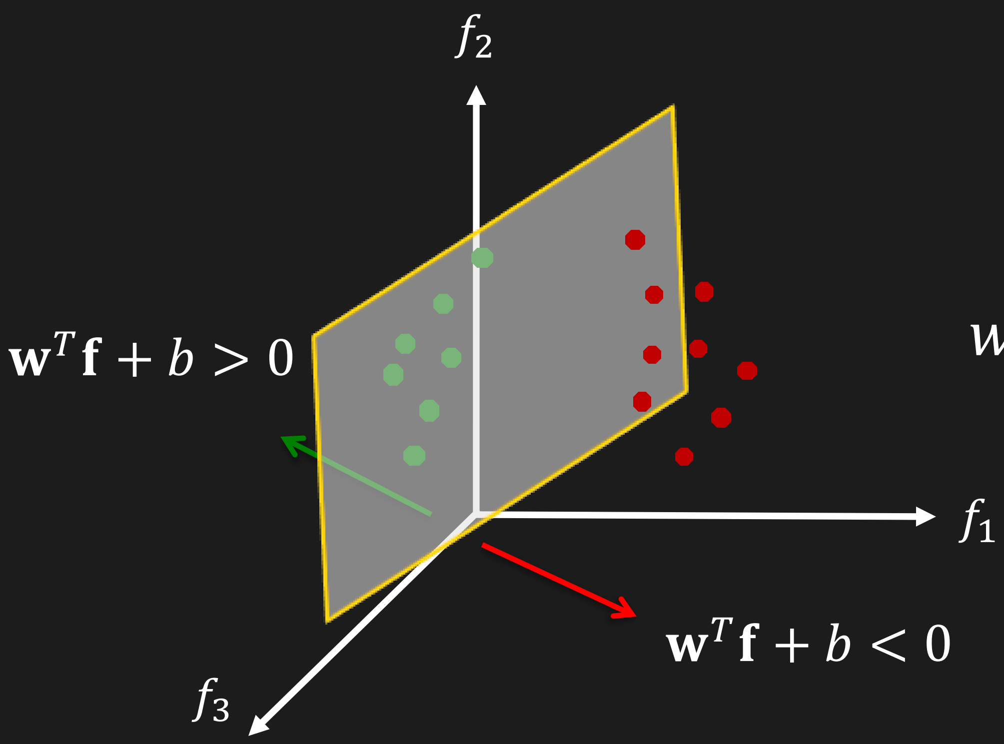
$$w_1 f_1 + w_2 f_2 + b = 0$$

$$\begin{bmatrix} w_1 & w_2 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} + b = 0$$

$$\boxed{\mathbf{w}^T\mathbf{f} + b = 0}$$

Linear Decision Boundaries

A Linear Decision Boundary in 3-D space is a 2-D Plane



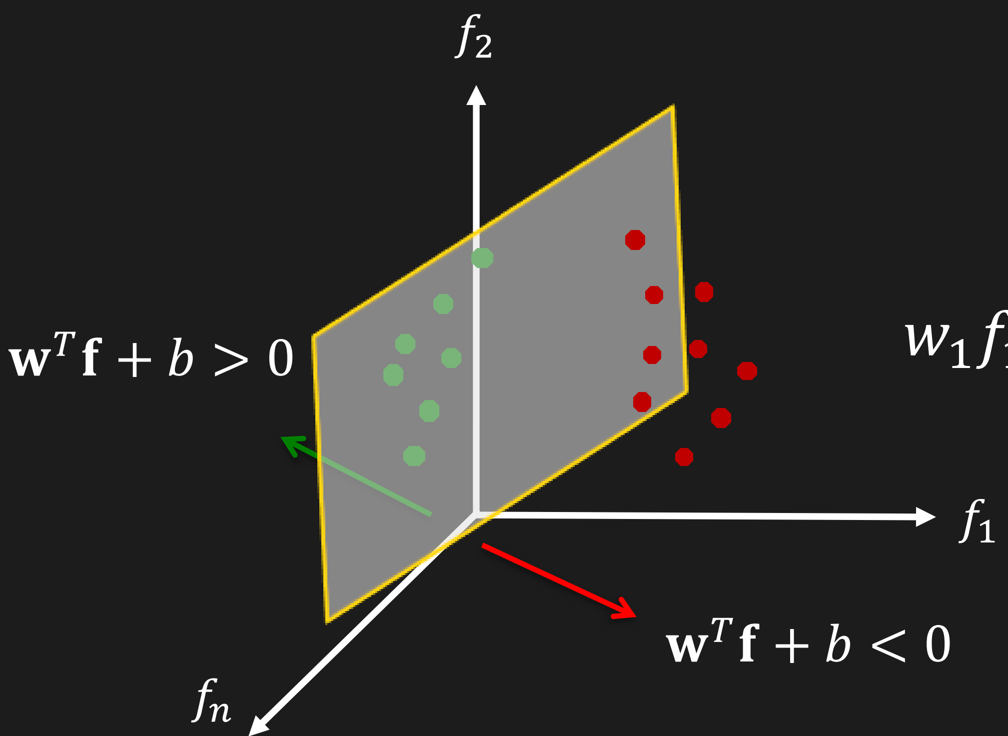
Equation of Plane:

$$w_1 f_1 + w_2 f_2 + w_3 f_3 + b = 0$$

$$\mathbf{w}^T\mathbf{f} + b = 0$$

Linear Decision Boundaries

A Linear Decision Boundary in n-D space
is a $(n-1)$-D Hyperplane



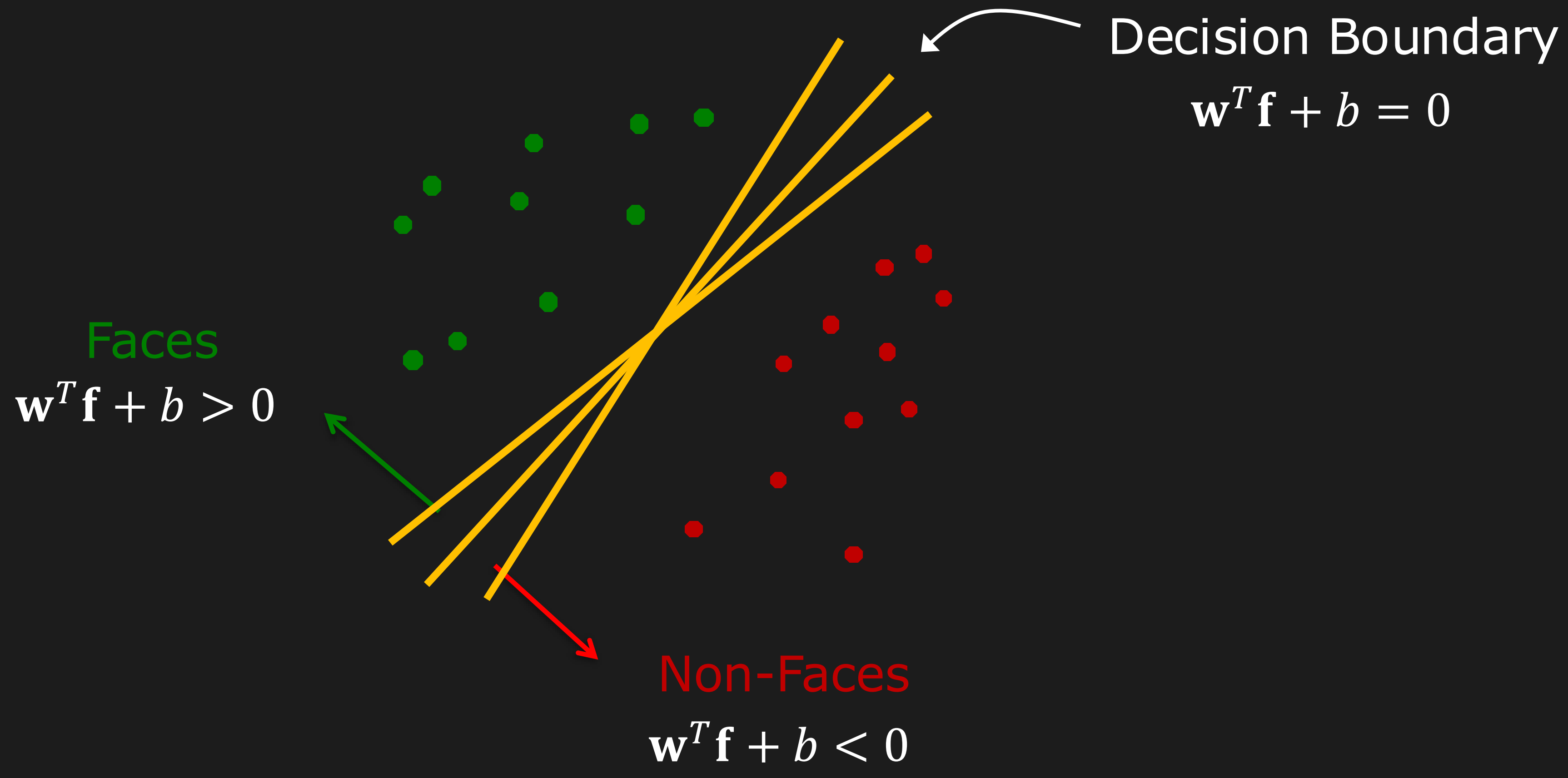
Equation of Hyperplane:

$$w_1 f_1 + w_2 f_2 + \cdots + w_n f_n + b = 0$$

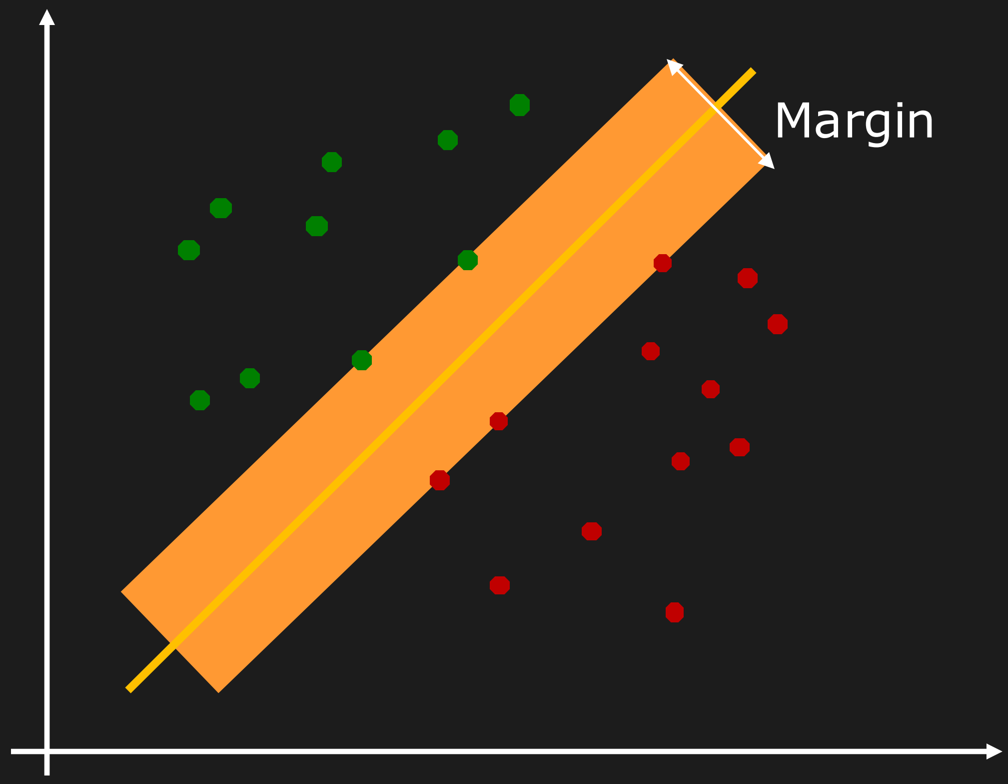
$$\mathbf{w}^T\mathbf{f} + b = 0$$

Decision Boundary $(\mathbf{w}, b)$

What is the optimal decision boundary?

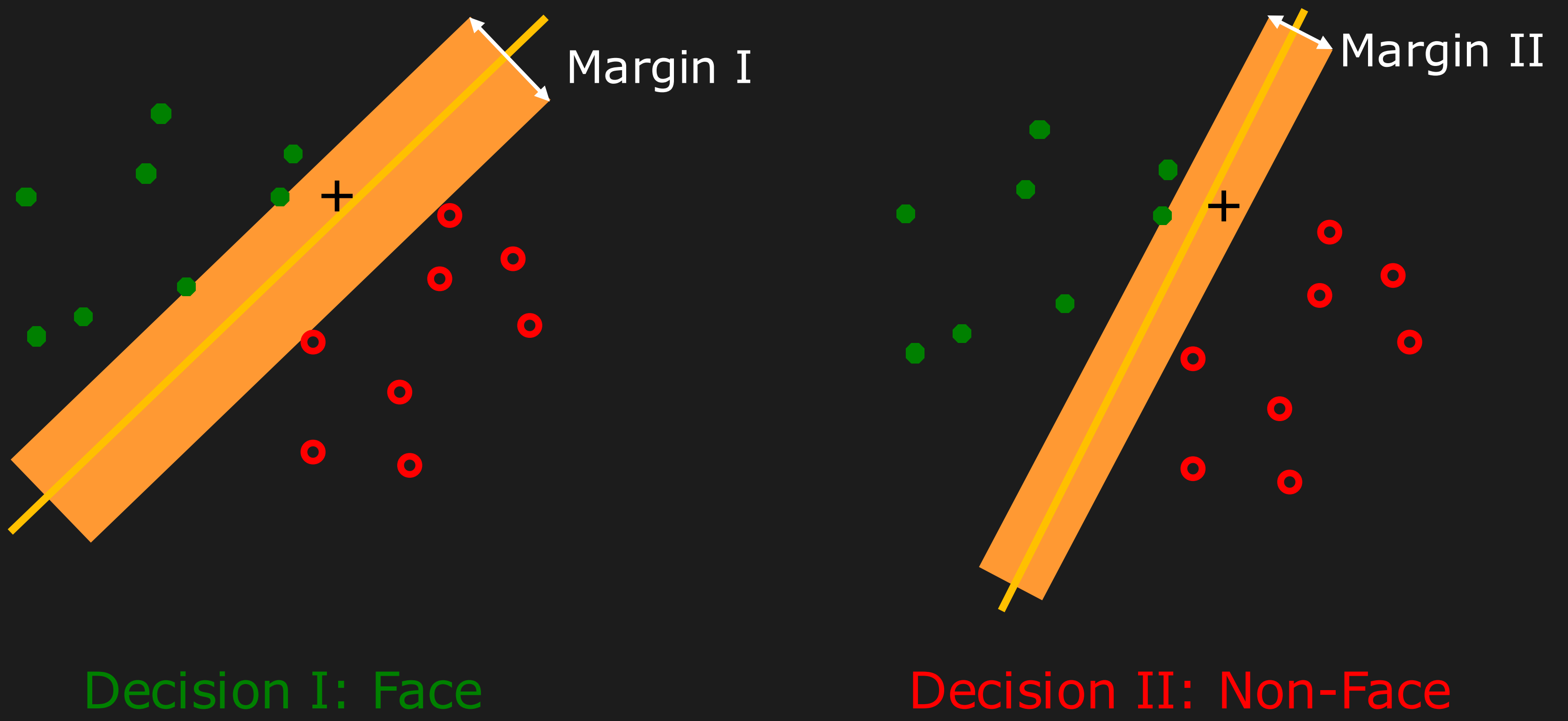


Evaluating a Decision Boundary



Margin or Safe Zone: The width that the boundary could be increased by, before hitting a feature point.

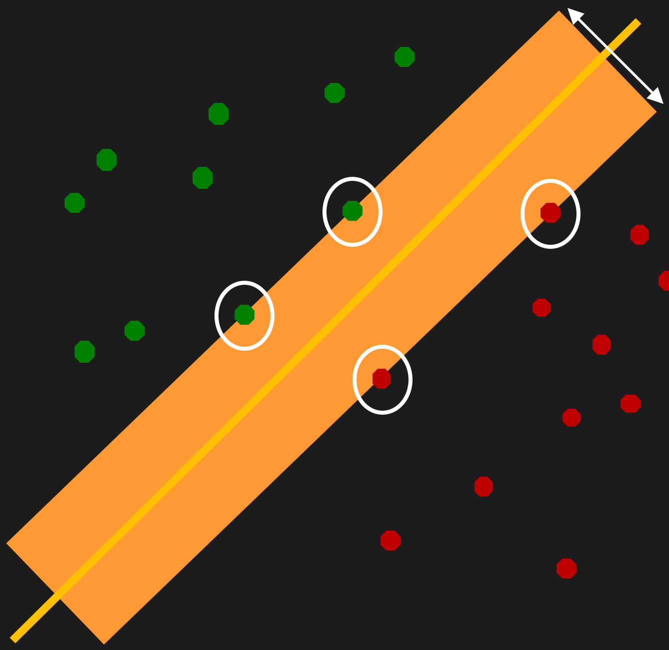
Evaluating a Decision Boundary



Choose Decision Boundary with Maximum Margin!

Support Vector Machine (SVM)

Classifier optimized to Maximize Margin



Support Vectors: Closest data samples to the boundary.

Decision Boundary and the Margin depend only on the Support Vectors.

[Cortes 1995]

Support Vector Machine (SVM)

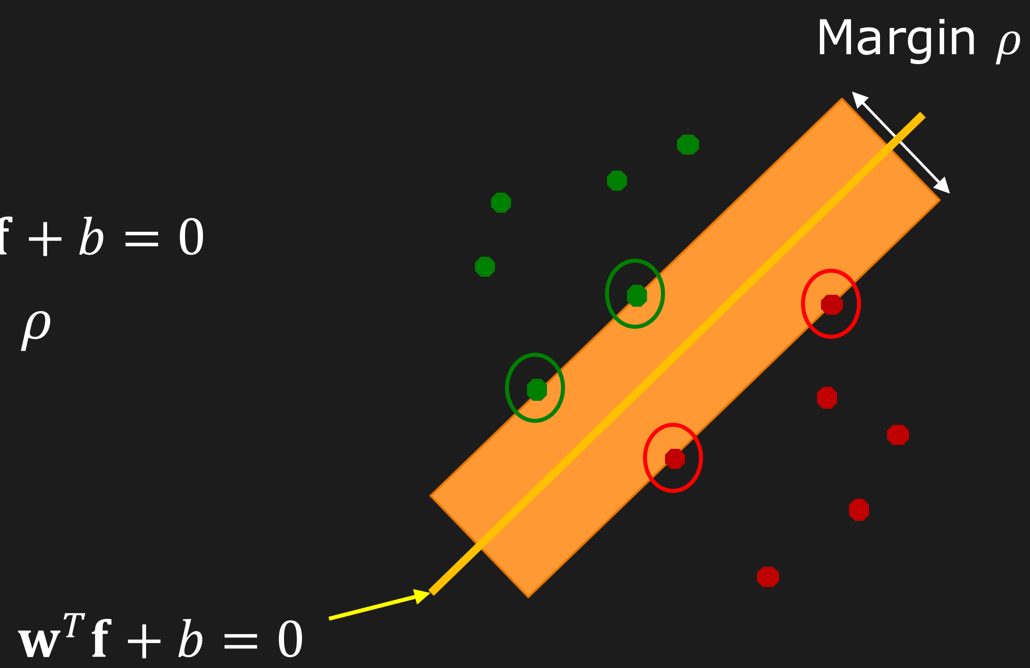
Given:

- k training images $\{I_1, I_2, \dots, I_k\}$ and their Haar features $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_k\}$.
- k corresponding labels $\{\lambda_1, \lambda_2, \dots, \lambda_k\}$, where $\lambda_j = +1$ if I_j is a face and $\lambda_j = -1$ if I_j is not a face.

Find:

Decision Boundary $\mathbf{w}^T\mathbf{f} + b = 0$

with Maximum Margin ρ

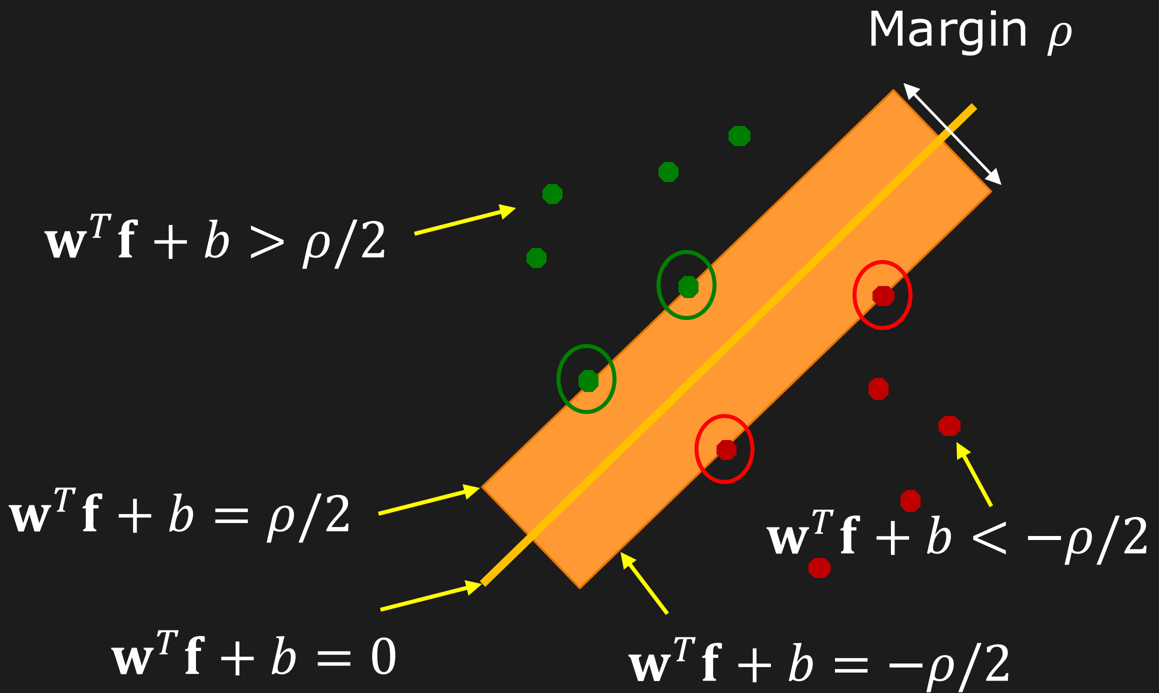


[Cortes 1995]

Finding Decision Boundary $(\boldsymbol{W}, b)$

For each training sample $(\mathbf{f}_i, \lambda_i)$:

$$\left.\begin{array}{ll} \text{If } \lambda_i = +1: & \mathbf{w}^T\mathbf{f}_i + b \geq \rho/2 \\ \text{If } \lambda_i = -1: & \mathbf{w}^T\mathbf{f}_i + b \leq -\rho/2 \end{array}\right\} \boxed{\lambda_i(\mathbf{w}^T\mathbf{f}_i + b) \geq \rho/2}$$



Finding Decision Boundary $(\boldsymbol{W}, b)$

For each training sample $(\mathbf{f}_i, \lambda_i)$:

$$\left.\begin{array}{ll} \text{If } \lambda_i = +1: & \mathbf{w}^T\mathbf{f}_i + b \geq \rho/2 \\ \text{If } \lambda_i = -1: & \mathbf{w}^T\mathbf{f}_i + b \leq -\rho/2 \end{array}\right\} \boxed{\lambda_i(\mathbf{w}^T\mathbf{f}_i + b) \geq \rho/2}$$

If $\mathcal{S}$ is the set of support vectors,

Then for every support vector $s \in \mathcal{S}$: $\boxed{\lambda_s(\mathbf{w}^T\mathbf{f}_s + b) = \rho/2}$

Numerical methods exist to find
$\mathbf{w}, b$ and $\mathcal{S}$ that maximize ρ

MATLAB: `svmtrain`

[Cortes 1995]

Classification using SVM

Given: Haar features $\mathbf{f}$ at image window and

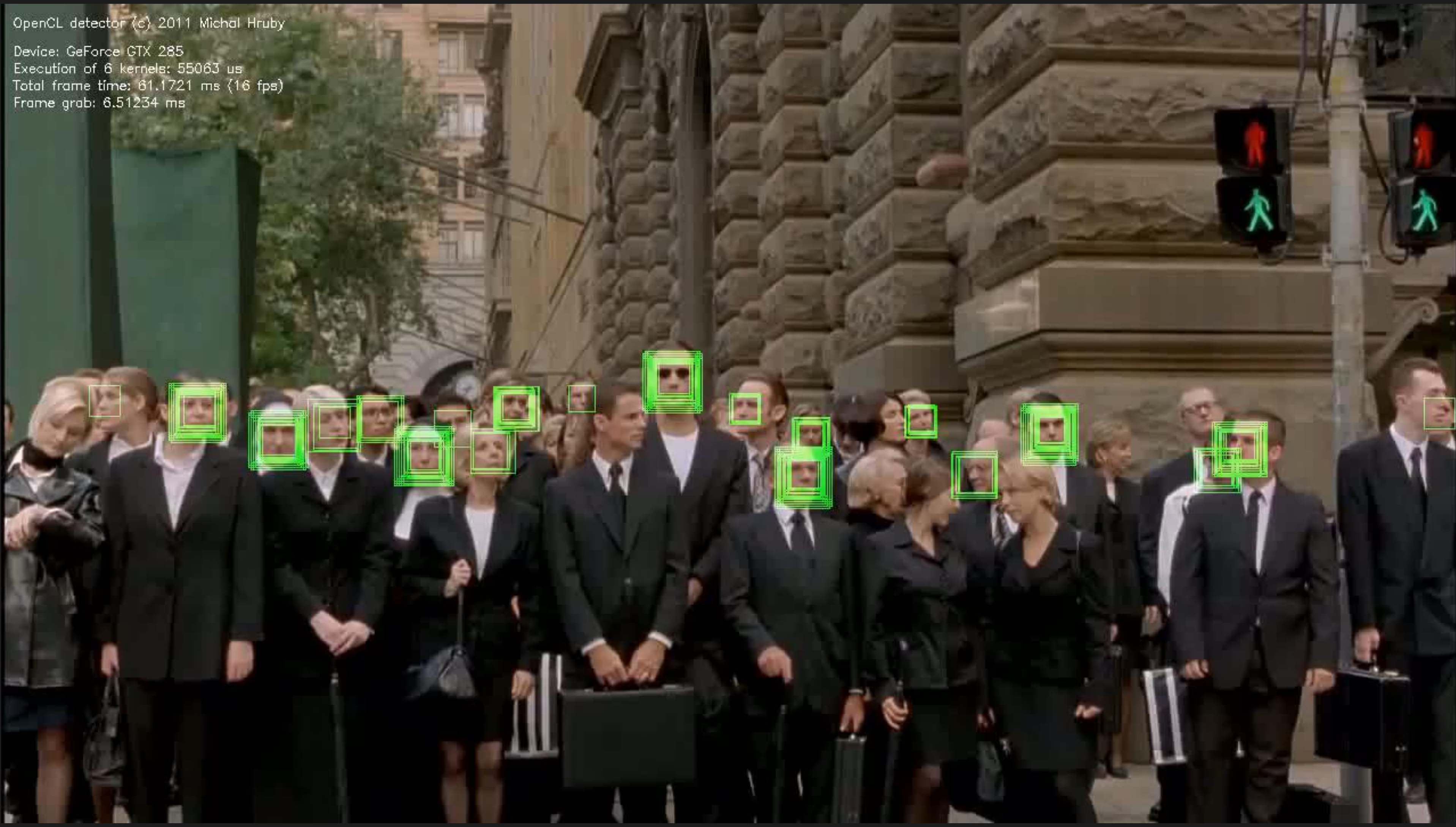
SVM parameters $\mathbf{w}, b, \rho$

Classification:

Compute $d = \mathbf{w}^T \mathbf{f} + b$

If: $\begin{cases} d \geq \rho/2 & \text{Face} \\ d > 0 \; and \; d < \rho/2 & \text{Probably Face} \\ d < 0 \; and \; d > -\rho/2 & \text{Probably Not-Face} \\ d \leq -\rho/2 & \text{Not-Face} \end{cases}$

Face Detection Results



Comments

- Current face detection systems mature but not perfect
- Frontal and side poses usually require different face models
- Successful vision technology used in cameras, surveillance, biometrics, search.
- Performance continues to improve.

References: Papers

[Burges 1998] C. J. C. Burges. “A Tutorial on Support Vector Machines for Pattern Recognition”. 1998.

[Cortes 1995] C. Cortes and V. N. Vapnik. "Support-Vector Networks". Machine Learning 1995.

[Crow 1985] F. Crow. "Summed-area tables for texture mapping" . SIGGRAPH 1984.

[Freund 1996] Y. Freund and R. E. Schapire. “Experiments with a new boosting algorithm”. 1996.

[Viola 2004] P. Viola and M. Jones. “Robust real-time face detection”. 2004.

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